



Converting Between Systems

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Converting Between Systems

Your work with ratio and proportion can help you with more complex measurement conversions. You can use **unit multipliers** to perform several conversions at once.

Examples of Unit Multipliers	$\frac{12 \text{ inches}}{1 \text{ foot}}$	$\frac{1 \text{ pound}}{16 \text{ ounces}}$	$\frac{100 \text{ centimeters}}{1 \text{ meter}}$

Can you see that each unit multiplier equals 1? In each case, the value in the numerator is equal to the value in the denominator.

Example

Hide

How many centimeters are in 3 kilometers?

Step 1

Write the measurement you are converting as a fraction over 1.

$$\frac{3 \text{ kilometers}}{1}$$

Step 2

Make a chain of unit multipliers using the facts that you know. You know that 1 km = 1,000 m and 100 cm = 1 m.

$$\frac{3 \text{ kilometers}}{1} \times \frac{1,000 \text{ meters}}{1 \text{ kilometer}} \times \frac{100 \text{ centimeters}}{1 \text{ meter}}$$

Step 3

Multiply the expression. You can cancel like measurement units just as you would cancel factors when multiplying fractions.

$$\frac{3 \cancel{\text{km}}}{1} \times \frac{1,000 \cancel{\text{m}}}{1 \cancel{\text{km}}} \times \frac{100 \text{cm}}{1 \cancel{\text{m}}} = \frac{3 \times 1,000 \times 100 \text{ cm}}{1} = \mathbf{300,000 \text{ cm}}$$

There are **300,000 centimeters** in 3 kilometers.

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Category
Lesson 2: The Metric System

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